Chapter 5

Segmentation

Region Splitting

➢Region Based Segmentation

- A region can be classified as a group of connected pixels exhibiting similar properties.
- The similarity between pixels can be in terms of intensity, color, etc.
- Region-based segmentation methods are preferred over edge-based segmentation methods in case of a noisy image.
- Region-Based techniques are further classified into 2 types based on the approaches they follow.
  - Region growing method
  - Region splitting and merging method

Region Splitting

➢Overview of Region Growing

- Start with some pixel as the seed pixel and then check the adjacent pixels.
- If the adjacent pixelsabide by the predefined rules, then that pixel is added to the region of the seed pixel and
- the process continues till there is no similarity left. This method follows the bottom-up approach.
- In case of a region growing, the preferred rule can be set as a threshold.

Region Splitting

➢Example

- Consider a seed pixel of 2 in the given image and a threshold value of 3,
- if a pixel has a value greater than 3 then it will be considered inside the seed
- pixel region. Otherwise, it will be considered in another region.

1	1	5	6	5	5
2	1	6	7	4	6
3	2	7	4	6	7
1	0	5	5	7	6
2	0	4	6	8	5
0	1	6	4	5	8

Original Image

1	1	5	6	5	5
2	1	6	7	4	6
3	2	7	4	6	7
1	0	5	5	7	6
2	0	4	6	8	5
0	1	6	4	5	8

Region growing process with 2 as the seed pixel.

R1	R1	R2	R2	R2	R2
R1	R1	R2	R2	R2	R2
R1	R1	R2	R2	R2	R2
R1	R1	R2	R2	R2	R2
R1	R1	R2	R2	R2	R2
R1	R1	R2	R2	R2	R2

Splitting image into two regions based on a threshold.

Region Splitting

➢Overview of Region Splitting

- Splitting the image into successively finer regions is one of the oldest techniques in computer vision.
- The basic idea of region splitting is to break the image into a set of disjoint regions which are coherent within themselves:
- This process is repeated until regions are either fairly uniform or below a certain size.

Region Splitting

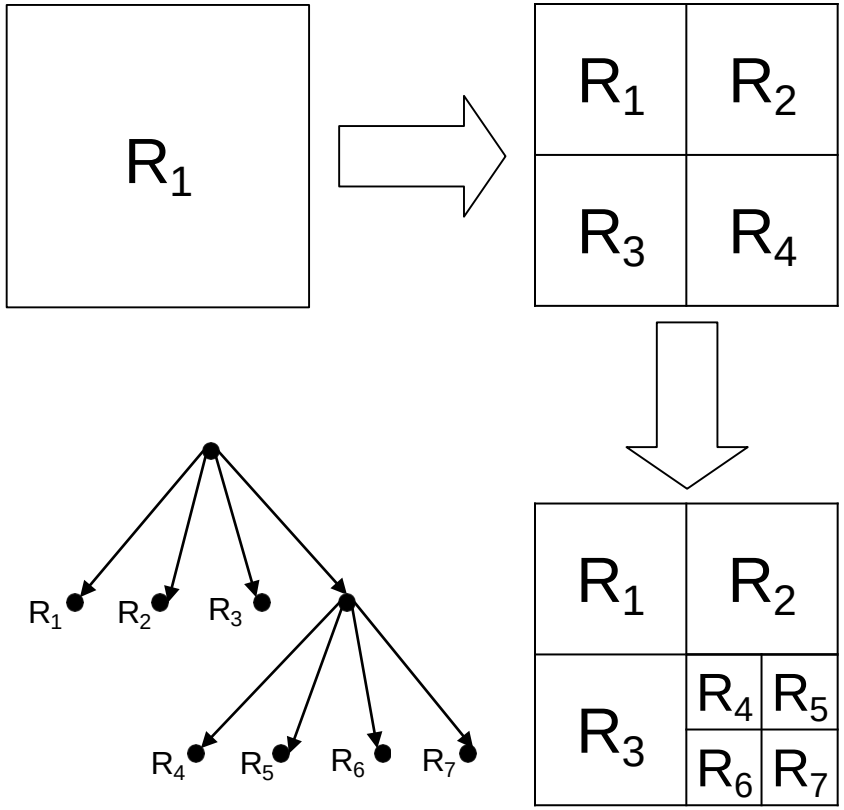
➢Overview of Region Merging

- If only a splitting schedule is used then the final segmentation would probably contain many neighbouring regions that have identical or similar properties.
- Thus, a merging process is used after each split which compares adjacent regions and merges them if necessary.
- We select a region as the seed region to check if adjacent regions are similar based on predefined rules.
- If they are similar, we merge them into a single region.

Region Splitting

➢Example(Splitting)

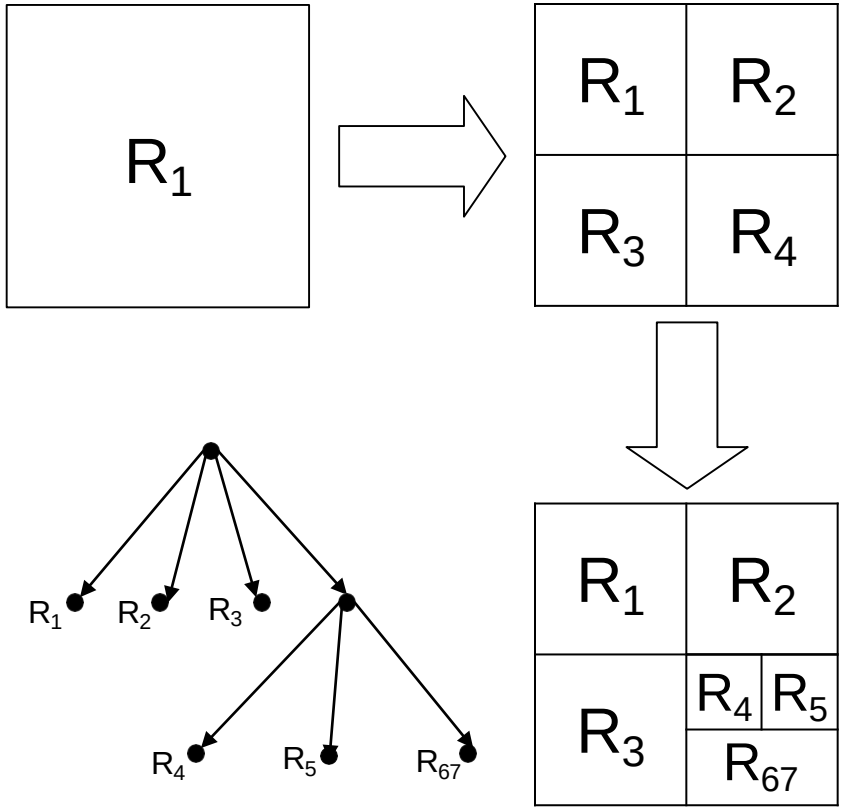
- One initial set that includes the whole image.
- Similarity criteria.
- Iteratively split regions into sub-regions.
- Stop when no more splitting are possible.



Region Splitting

➢Example(Merging)

- Now assume that all pixels within each region are similar with respect to that region
- after comparing the split regions, regions R6 and R7 are found to be identical.
- These are thus merged together.



Region Splitting

➢Region Splitting and Region Merging

- Both region splitting and region merging are iterative processes.
- Usually, first region splitting is done, an image.
- The image is successively split into quadrants based on a homogeneity criterion.
- And then similar regions are merged in order to form a good segmented image of the original image.
- Algorithms of this nature are called split and merge algorithm.

Region Splitting

➢Basic Algorithm

- Define the criterion to be used for homogeneity
- Split the image into equal size regions
- Calculate homogeneity for each region
- If the region is homogeneous, then merge it with neighbors
- The process is repeated until all regions pass the homogeneity test

Region Splitting

➢Split / Merge

- Splitting techniques disadvantage, they create regions that may be adjacent
- and homogeneous, but not merged.
- Split and Merge method – It is an iterative algorithm that includes both splitting and merging at each iteration:
- If a region R is **inhomogeneous** (P(R)= False) then is split into four sub regions
- If two adjacent regions **Ri,Rj are homogeneous** (P(Ri U Rj) = TRUE), they are merged
- The algorithm stops when no further splitting or merging is possible

Region Splitting

➢ Applications

- The split and merge algorithm produces more compact regions than the
- pure splitting algorithm
- 3D – Imaging : A basic task in 3-D image processing is the segmentation of an image which classifies voxels/pixels into objects or groups. .
- Several applications in the field of Medicine like magnetic resonance imaging (MRI).

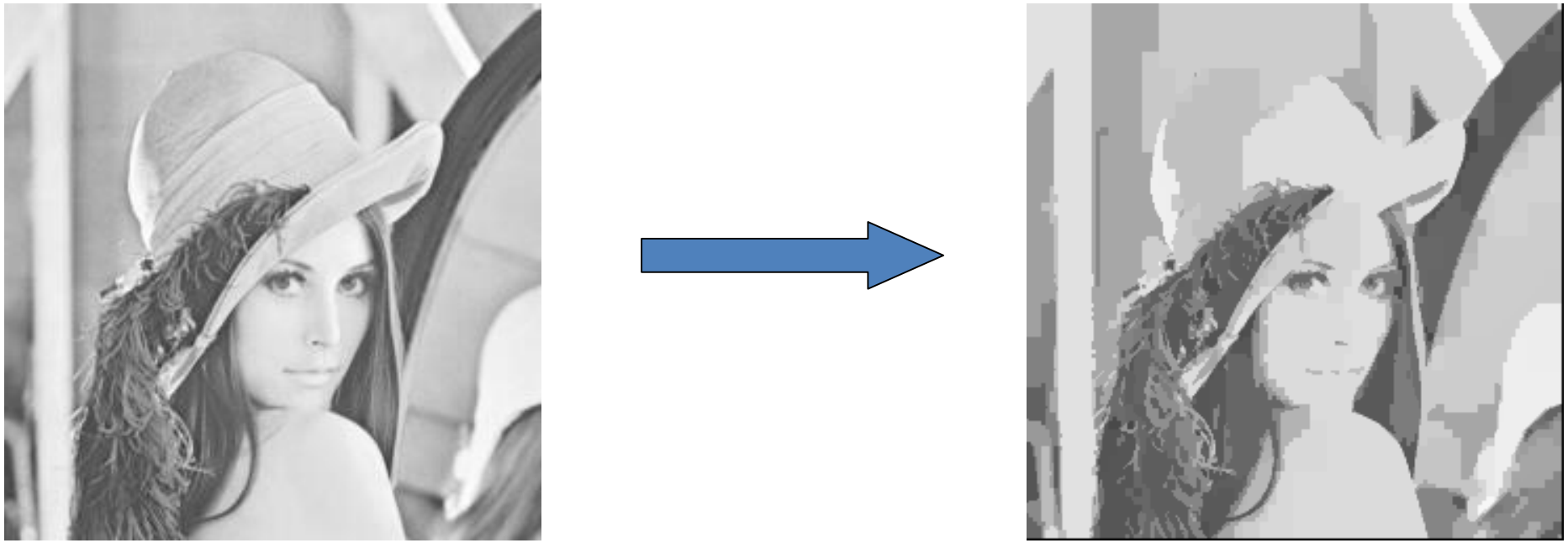
Region Splitting

- **Result – Region Split**

http://astro.temple.edu/~siddu

Region Splitting

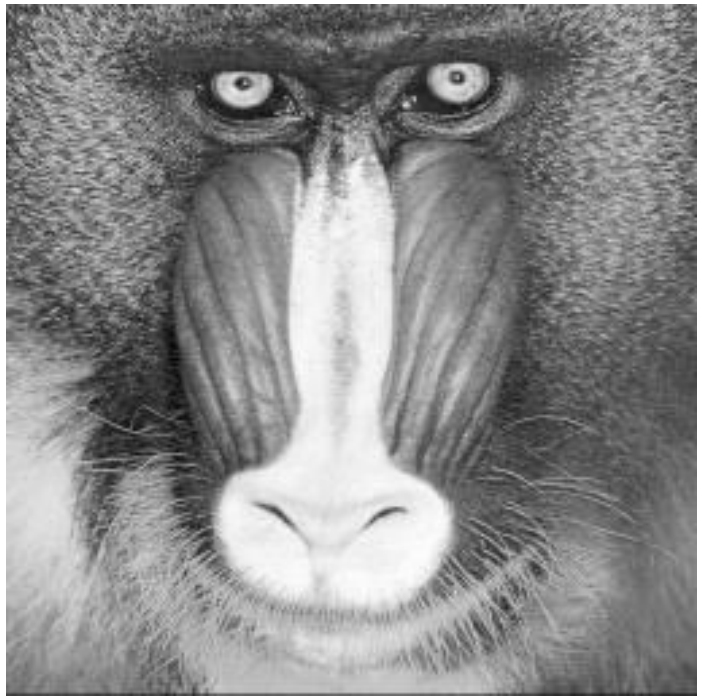
- **Result – Region Split and Merge**

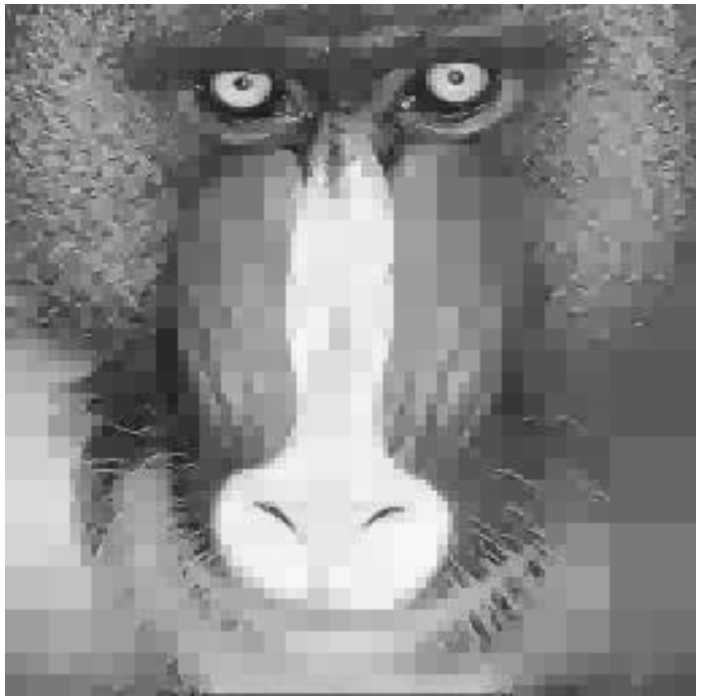


http://astro.temple.edu/~siddu

Region Splitting

- **Result – Region Split**

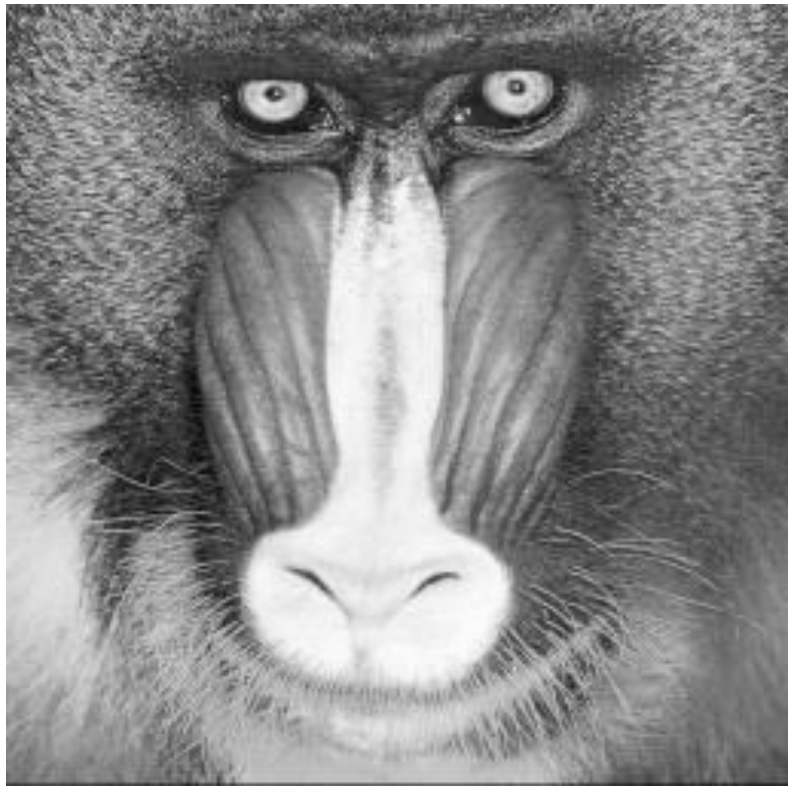


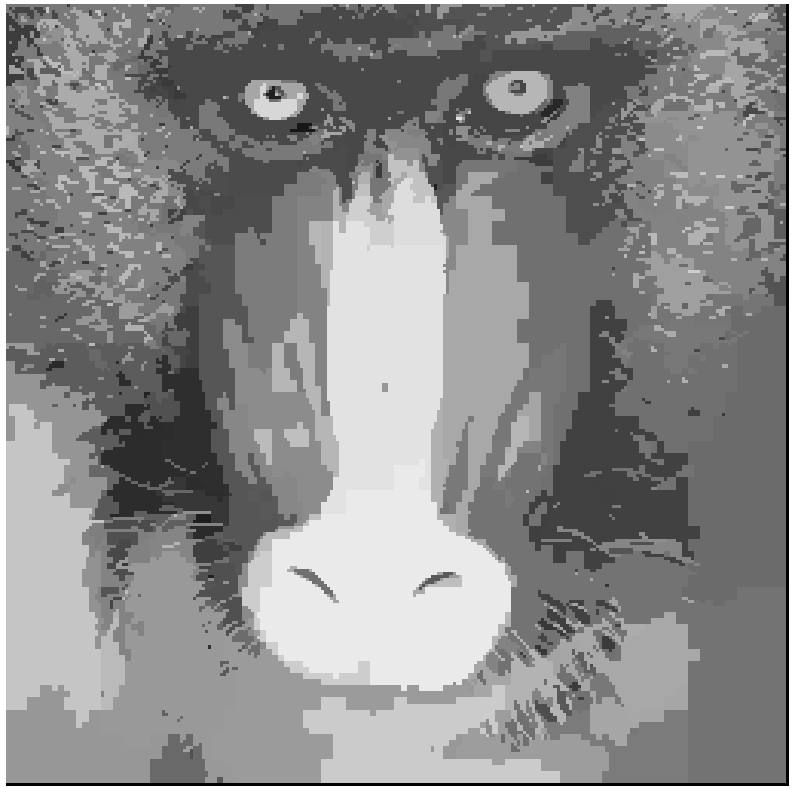


http://astro.temple.edu/~siddu

Region Splitting

- **Result – Region Split and Merge**





http://astro.temple.edu/~siddu

Region Splitting

➢ Question on Region Splitting

- Apply the region splitting and merging on the following image. Assume the threshold value be <=3.

5	6	6	6	7	7	6	6
6	7	6	7	5	5	4	7
6	6	4	4	3	2	5	6
5	4	5	4	2	3	4	6
0	3	2	3	3	2	4	7
0	0	0	0	2	2	5	6
1	1	0	1	0	3	4	4
0	2	1	0	2	3	5	4

https://www.youtube.com/watch?v=mn3n D3bEnO0

Region Splitting

➢ **Solution:**

- If the absolute value of the difference of the maximum and minimum pixel intensities in a region is less than or equal to a threshold value
  - then the region does not require further splitting.

$$Z\,\max - Zmin \quad \leq threshold$$

Zmax = Maximum pixel intensity value in a region.

Zmin = Minimum pixel intensity value in a region

Region Splitting

- **Solution: Splitting**

	5	6	6	6	7	7	6	6	
	6	7	6	7	5	5	4	7	
a	6	6	4	4	3	2	5	6	**b**
	5	4	5	4	2	3	4	6	
	0	3	2	3	3	2	4	7	
c	0	0	0	0	2	2	5	6	**d**
	1	1	0	1	0	3	4	4	
	0	2	1	0	2	3	5	4	

7 – 0 = 7

Region Splitting

- **Solution: Splitting**

7 – 4 = 3 **6 – 1 = 5**

	5	6	6	6	7	7	6	6
	6	7	6	7	5	5	4	7
R1	6	6	4	4	3	2	5	6 **R2**
	5	4	5	4	2	3	4	6
	0	3	2	3	3	2	4	7
R3	0	0	0	0	2	2	5	6 **R4**
	1	1	0	1	0	3	4	4
	0	2	1	0	2	3	5	4

3 – 1 = 2 **7 – 7 = 0**

Region Splitting

- **Solution: Splitting**

	5	6	6	6	7	7	6	6
	6	7	6	7	5	5	4	7
a	6	6	4	4	3	2	5	6
	5	4	5	4	2	3	4	6
	0	3	2	3	3	2	4	7
	0	0	0	0	2	2	5	6
c	1	1	0	1	0	3	4	4
	0	2	1	0	2	3	5	4

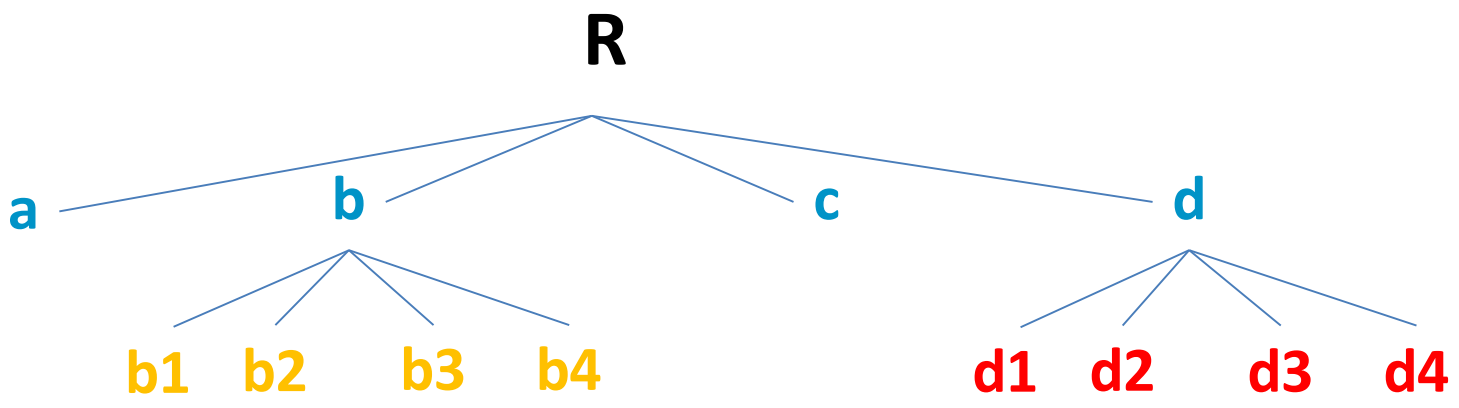
b1 **b2** **b3** **b4** **d1** **d2** **d3** **d4**

b **d**

Region Splitting

➢ Solution: Splitting (Region Quadtree)

- The region quadtree represents a partition of space in two dimensions by decomposing the region into four equal quadrants, subquadrants, and so on with each leaf node containing data corresponding to a specific subregion.
- Each node in the tree either has exactly four children, or has no children (a leaf node).



Region Splitting

➢ Solution: Merging

- Check the adjacent regions, if they are falling within the threshold, then merge.

	5	6	6	6	7	7	6	6
	6	7	6	7	5	5	4	7
a	6	6	4	4	3	2	5	6
	5	4	5	4	2	3	4	6
	0	3	2	3	3	2	4	7
	0	0	0	0	2	2	5	6
c	1	1	0	1	0	3	4	4
	0	2	1	0	2	3	5	4

b1 b2 b3 b4 d1 d2 d3 d4

b

d

Region Splitting

- **Solution: Merging**

- Check the adjacent regions, if they are falling within the threshold, then merge

Consider region a and b1

Max = 7, Min = 4

| 7 – 4 | = 3 <= 3 (Merge)

Consider region a-b1 and b2

Max = 7, Min = 4

| 7 – 4 | = 3 <= 3 (Merge)

Consider region a-b1-b2 and b4

Max = 7, Min = 4

| 7 – 4 | = 3 <= 3 (Merge)

Consider region a-b1-b2-b4 and d2

Max = 7, Min = 4

| 7 – 4 | = 3 <= 3 (Merge)

Consider region a-b1-b2-b4-d2 and d4

Max = 7, Min = 4

| 7 – 4 | = 3 <= 3 (Merge)

Region Splitting

➢ Solution: Merging

- Check the adjacent regions, if they are falling within the threshold, then merge.

	5	6	6	6	7	7	6	6
	6	7	6	7	5	5	4	7
a	6	6	4	4	3	2	5	6
	5	4	5	4	2	3	4	6
	0	3	2	3	3	2	4	7
c	0	0	0	0	2	2	5	6
	1	1	0	1	0	3	4	4
	0	2	1	0	2	3	5	4

b1 b2 b3 b4 d1 d2 d3 d4

b

d

Region Splitting

➢ Let's Summarize:

- Apply the region growing on the following image with initial point at (2, 2) and threshold value as 2. Use 4 connectivity.

0	1	2	0
2	5	6	1
1	4	7	3
0	2	5	1

7 – 6 = 1

7 – 5 = 2

7 – 4 = 3

7 – 3 = 4

Region Splitting

➢ Let's Summarize:

- Apply the region growing on the following image with initial point at (2, 2) and threshold value as 2. Use 4 connectivity.

0	1	2	0
2	5	6	1
1	4	7	3
0	2	5	1

6 – 2 = 4

6 – 5 = 1

6 – 1 = 5

0	1	2	0
2	5	6	1
1	4	7	3
0	2	5	1

5 – 2 = 3

5 – 1 = 4

0	0	0	0
0	1	1	0
0	0	1	0
0	0	1	0

Region Splitting

➢ Let's Summarize:

- Apply the region growing on the following image with seed point as 6 and threshold value as 3. Use 8 way connectivity.

5	6	6	7	6	7	6	6
6	7	6	7	5	5	4	7
6	6	4	4	3	2	5	6
5	4	5	4	2	3	4	6
0	3	2	3	3	2	4	7
0	0	0	0	2	2	5	6
1	1	0	1	0	3	4	4
1	0	1	0	2	3	5	4

Region Splitting

- **Let's Summarize:**

5	6	6	7	6	7	6	6
6	7	6	7	5	5	4	7
6	6	4	4	3	2	5	6
5	4	5	4	2	3	4	6
0	3	2	3	3	2	4	7
0	0	0	0	2	2	5	6
1	1	0	1	0	3	4	4
1	0	1	0	2	3	5	4

1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1
1	1	1	1	3	2	1	1
1	1	1	4	2	3	1	1
0	3	2	3	3	2	1	1
0	0	0	0	2	2	1	1
1	1	0	1	0	3	1	1
1	0	1	0	2	3	5	4

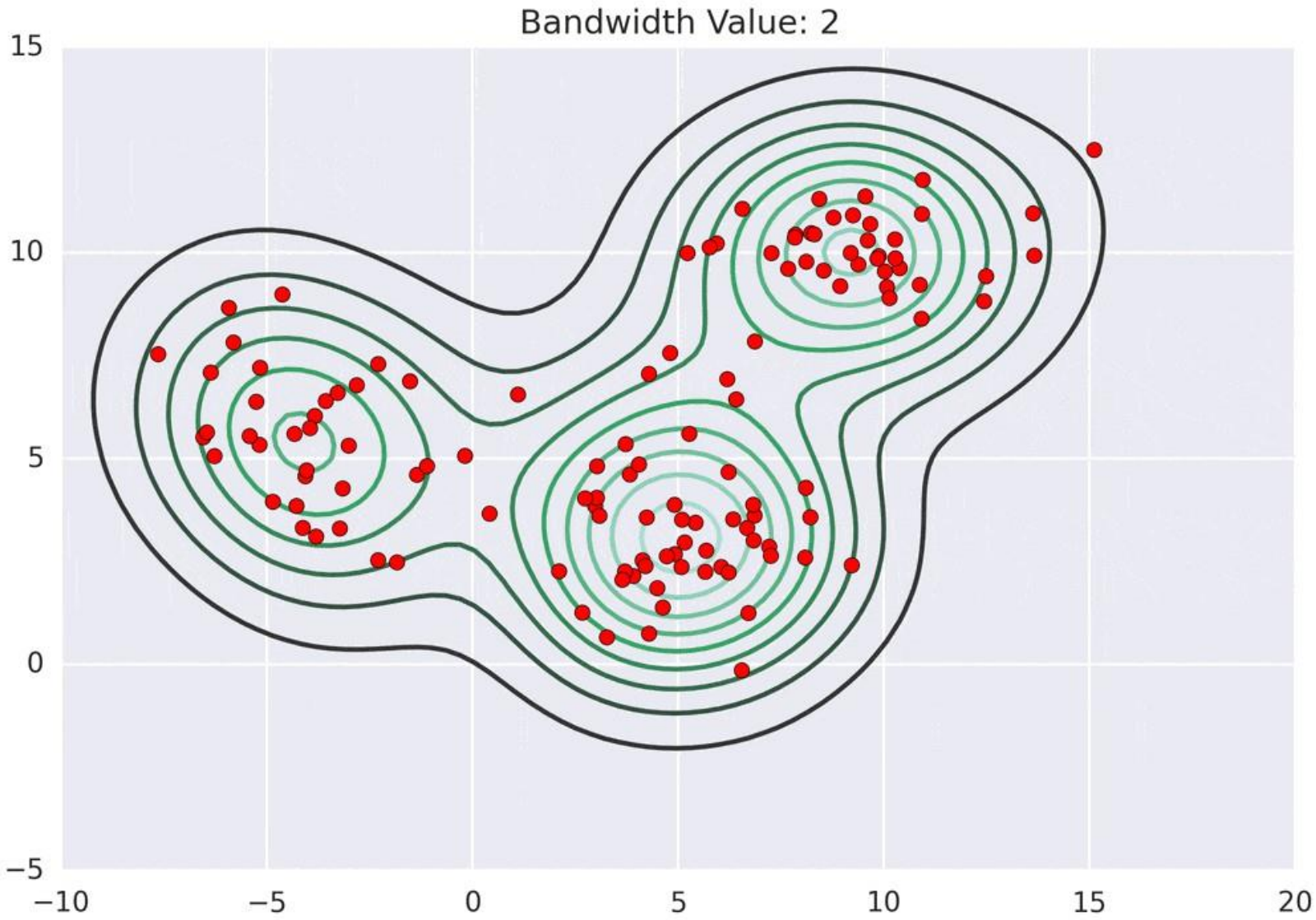
Region Splitting

➢ Kernel Density Estimation

- Mean shift is based on the concept of Kernel Density Estimation(KDE), which is a way to estimate the probability density function of a random variable.
- It works by providing weights to each data point. The weight function is called a kernel.
- There are many kinds of kernels, one kind of kernel is the Gaussian kernel.
- Adding all those kernels together creates a density function (probability surface).
- The resultant density function variation depends on the used bandwidth parameter.

Region Splitting

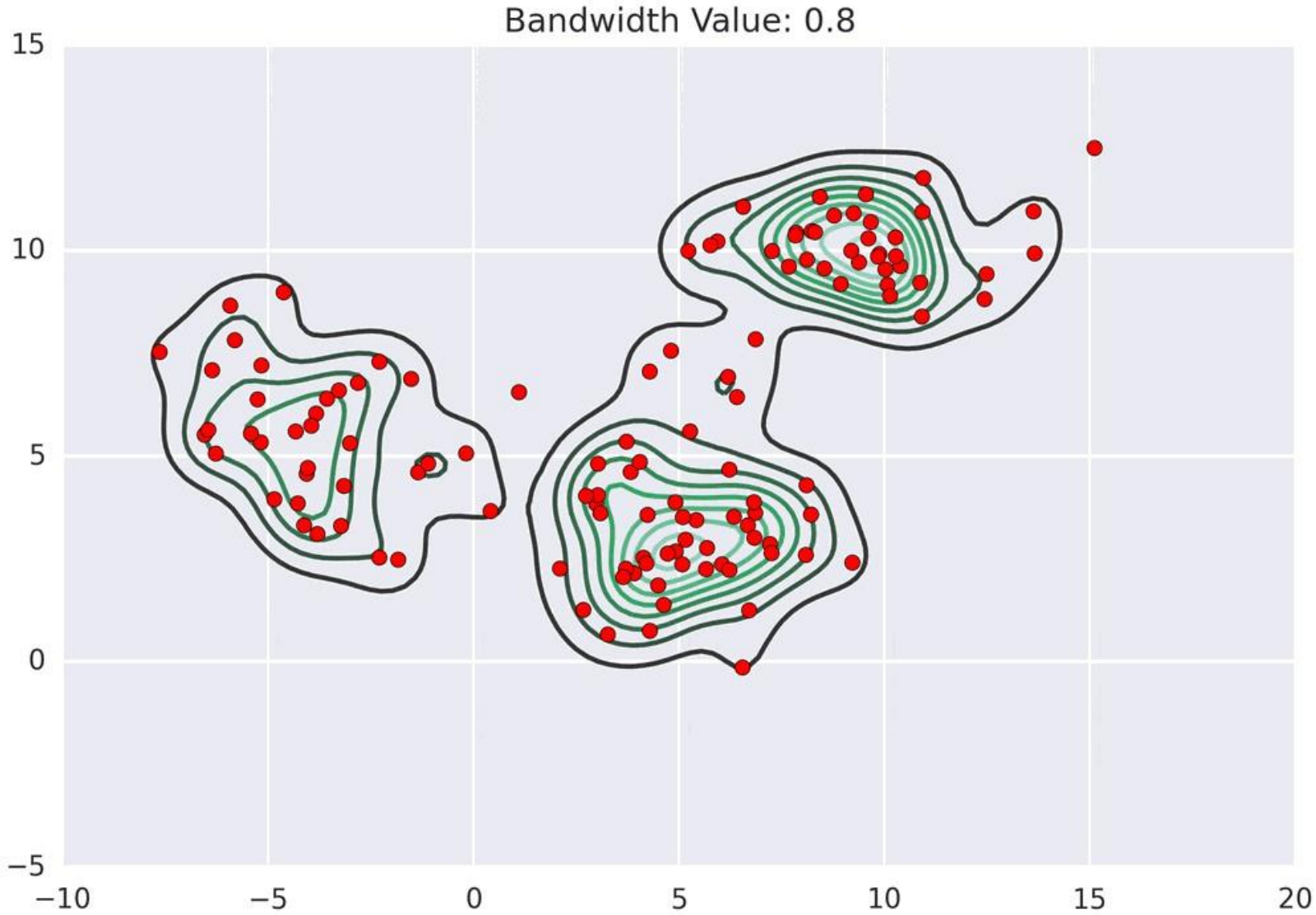
- **Bandwidth = 2**



https://analyticsindiamag.com/

Region Splitting

- **Bandwidth = 0.8**



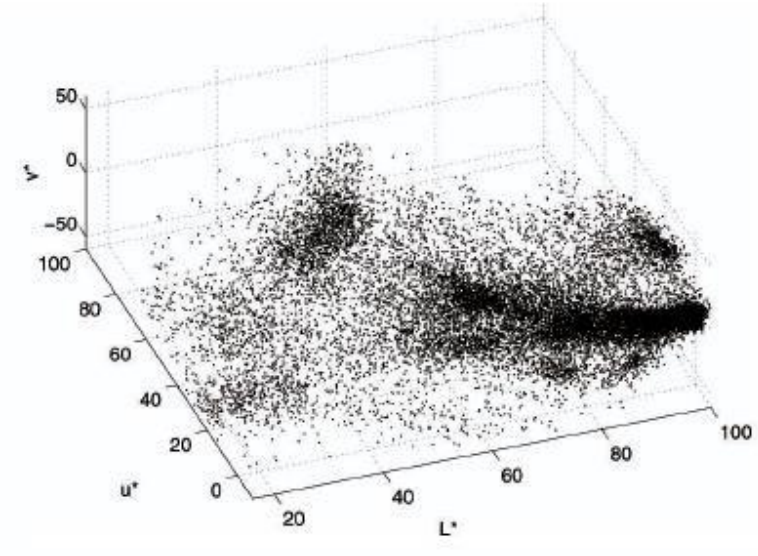
https://analyticsindiamag.com/

Region Splitting

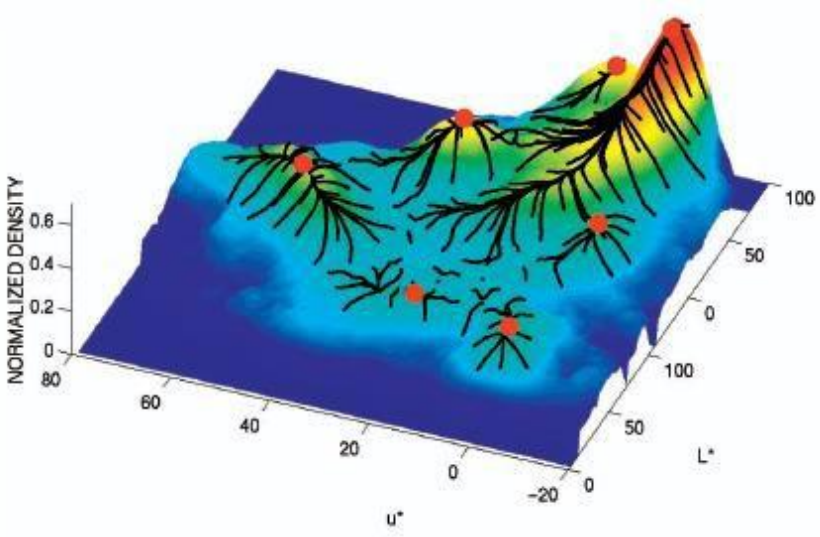
➢ Idea of Mean Shift

- Each hill represents a cluster.
- Peak (mode) of hill represents “center” of cluster.
- Each pixel climbs the steepest hill within its neighborhood.
- Pixel is assigned to the hill (cluster) it climbs.

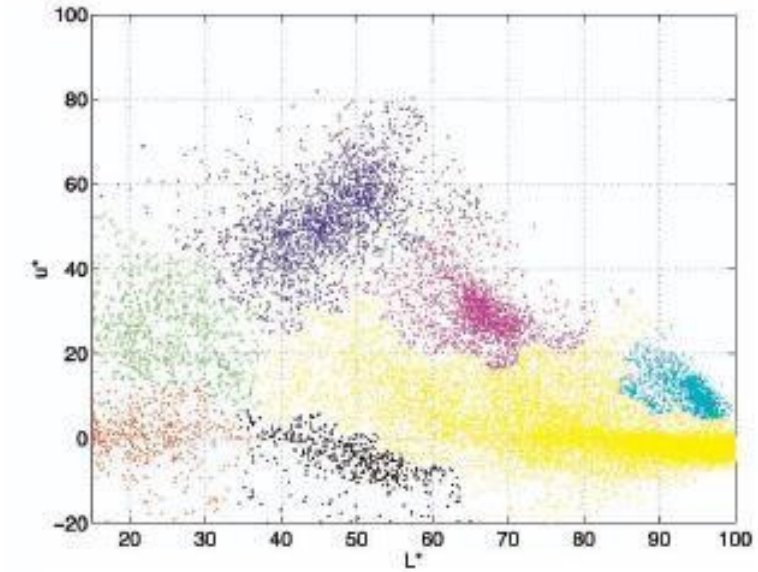
https://www.youtube.com/watch?v=PCNz_zttmtA



Pixel Feature Distribution



Normalize Density



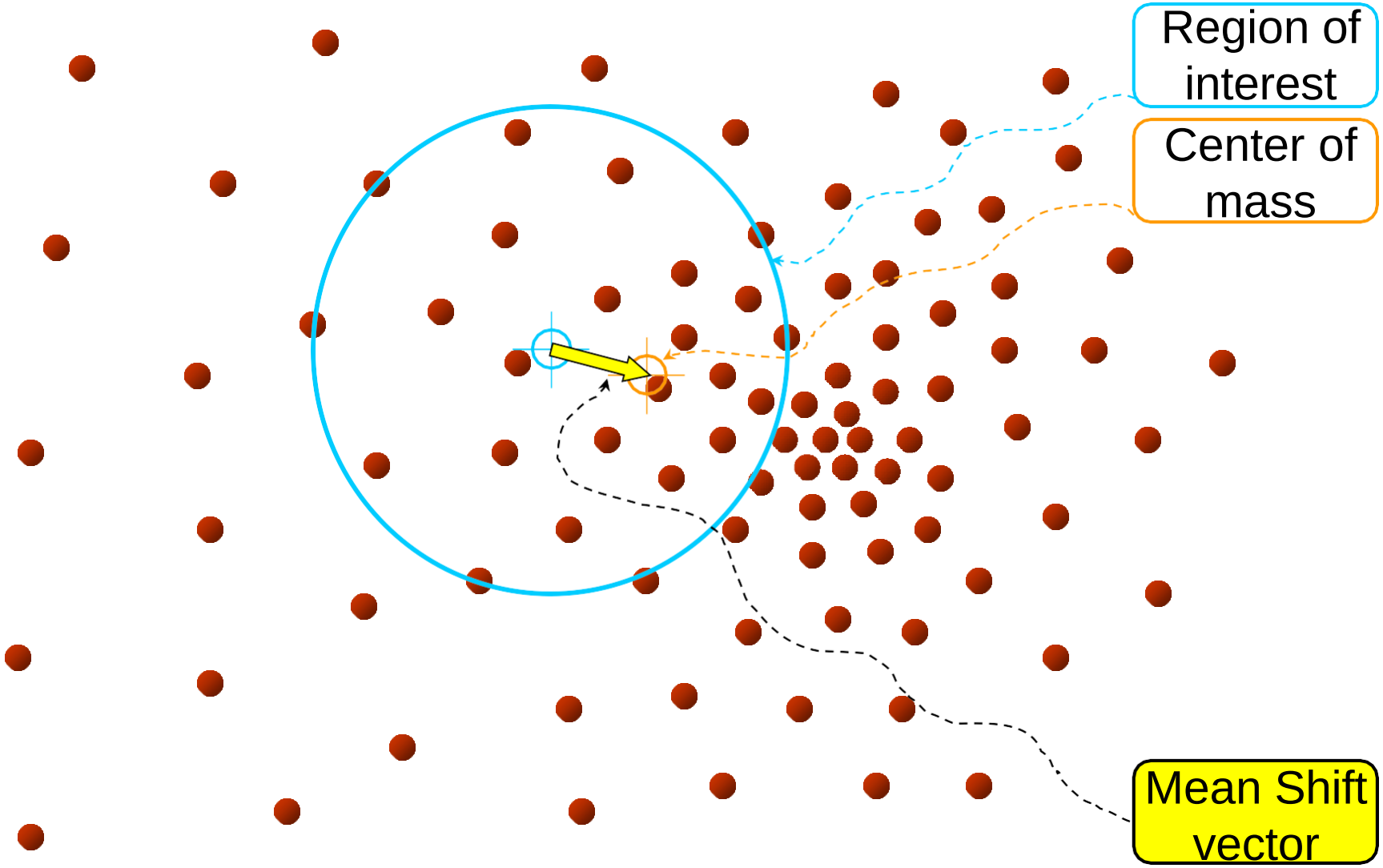
Label Clusters and their Centers

Region Splitting

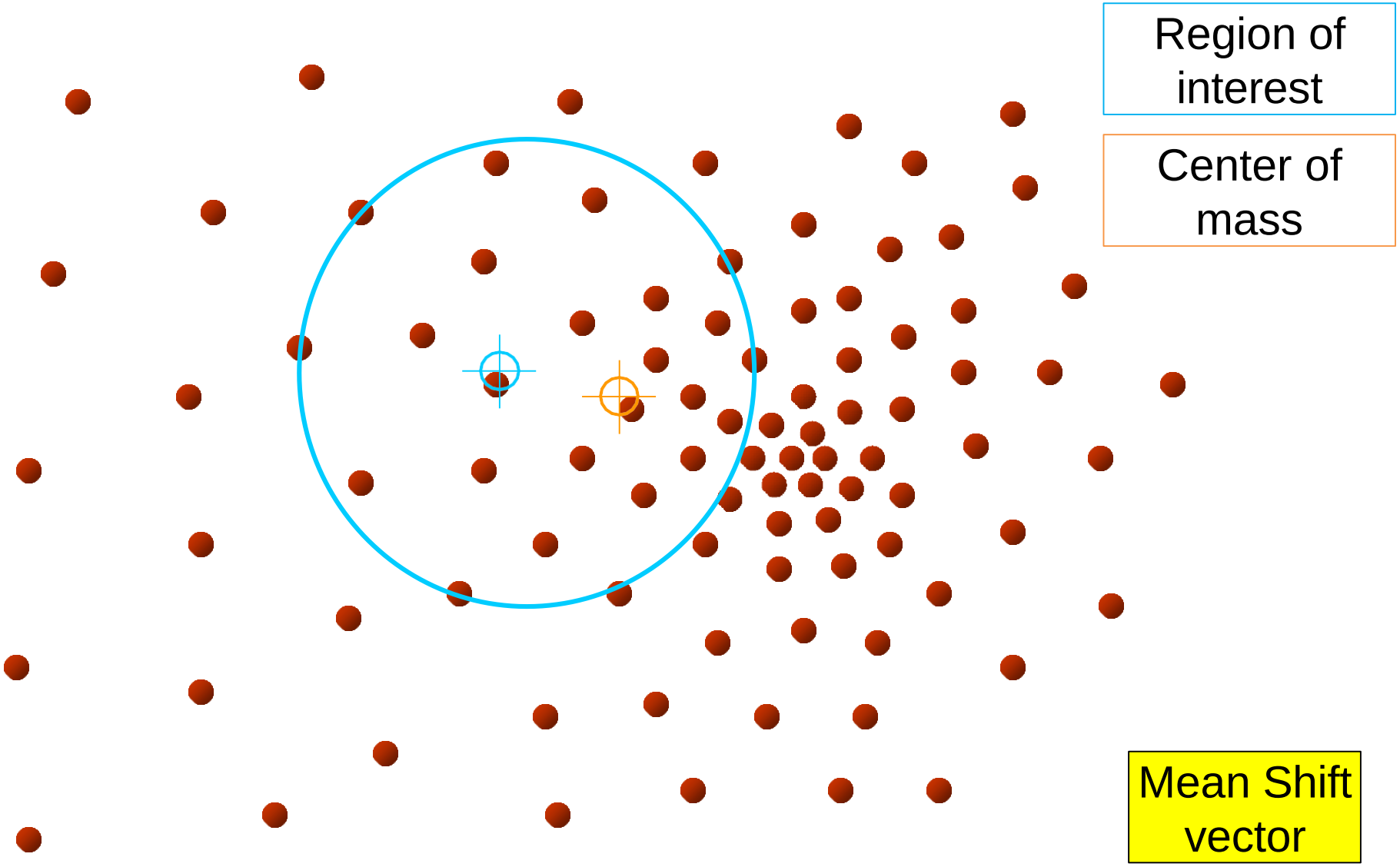
➢ Hill Climbing using Mean Shift

- Compute the centroid or mean within the Window size W.
- Use the Weighted Means.
- “Shift” the Window to the centroid.
- Repeat … Until Convergence.
- Declare mode and assign it a cluster label.
- Repeat for all pixel.
- Feature that converge to same mode belong to same cluster

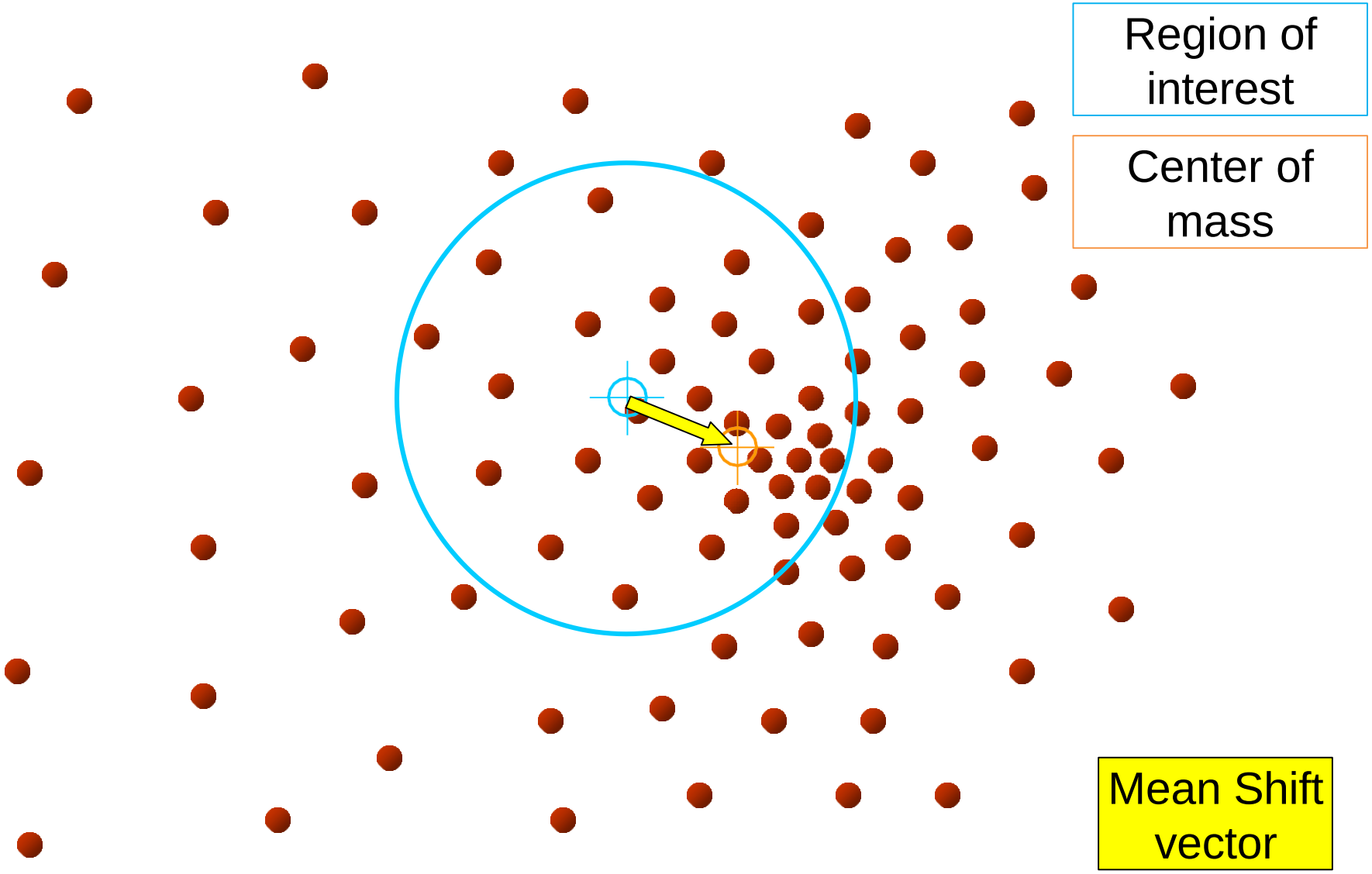
Region Splitting



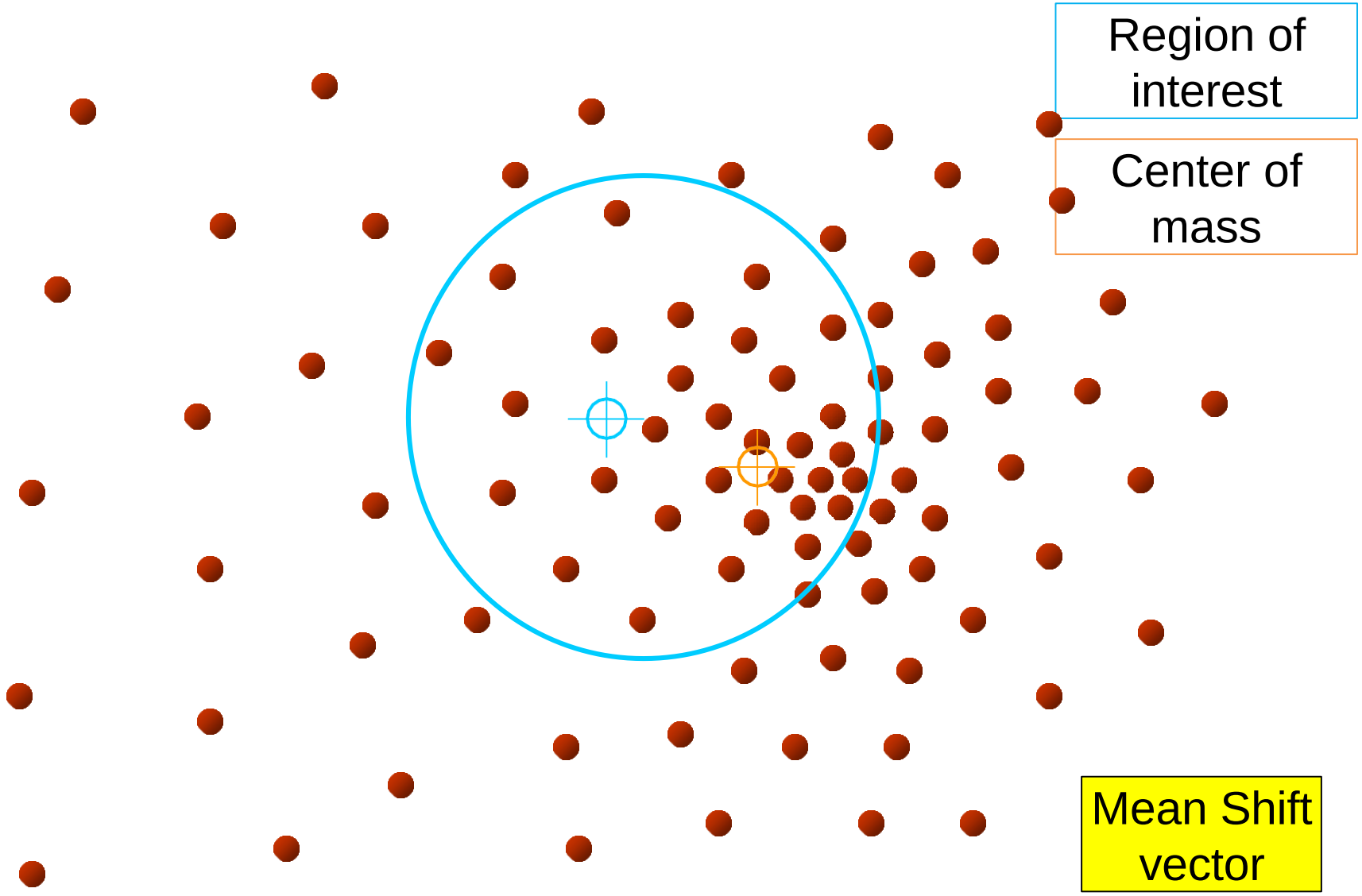
Region Splitting



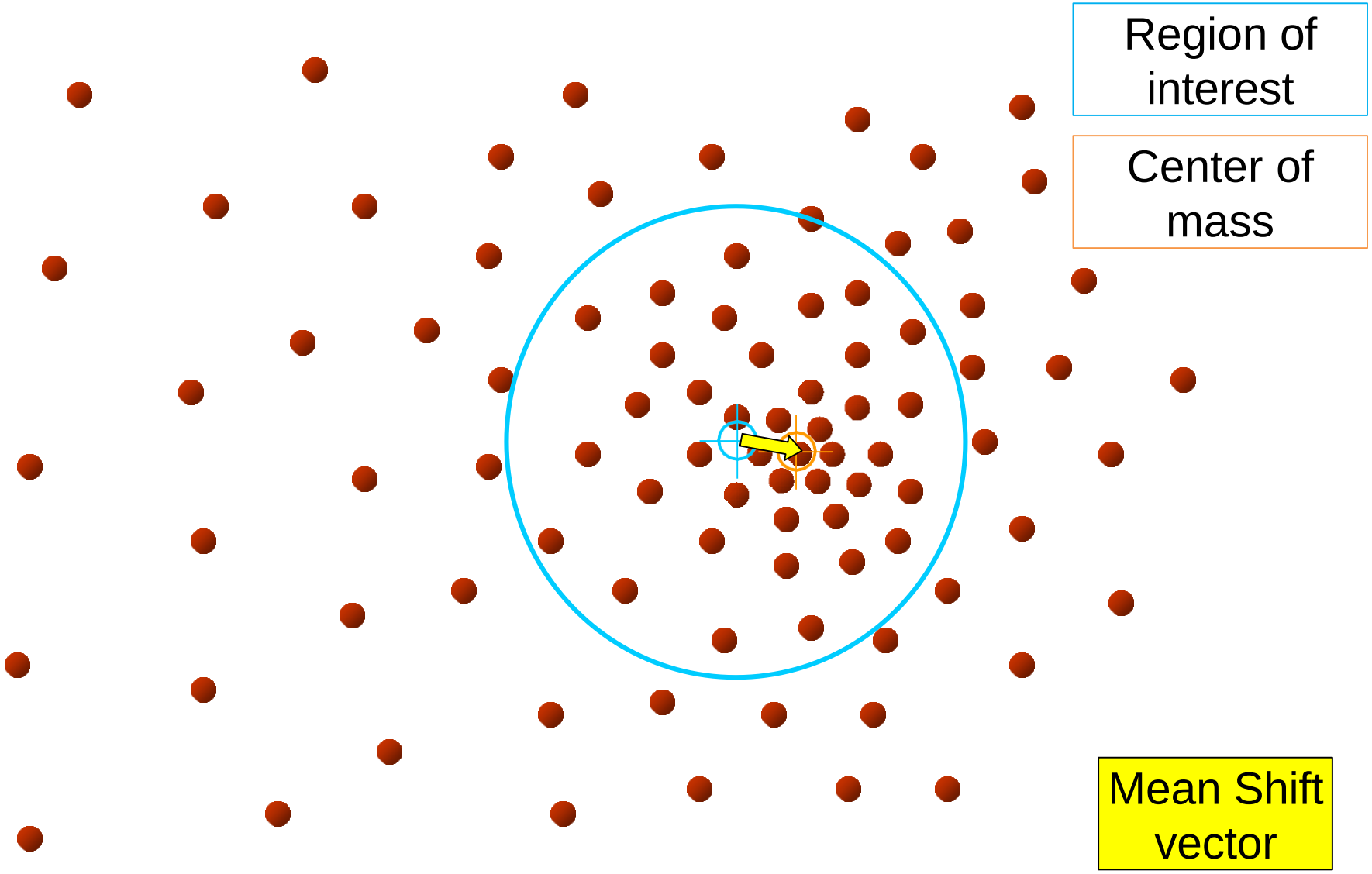
Region Splitting



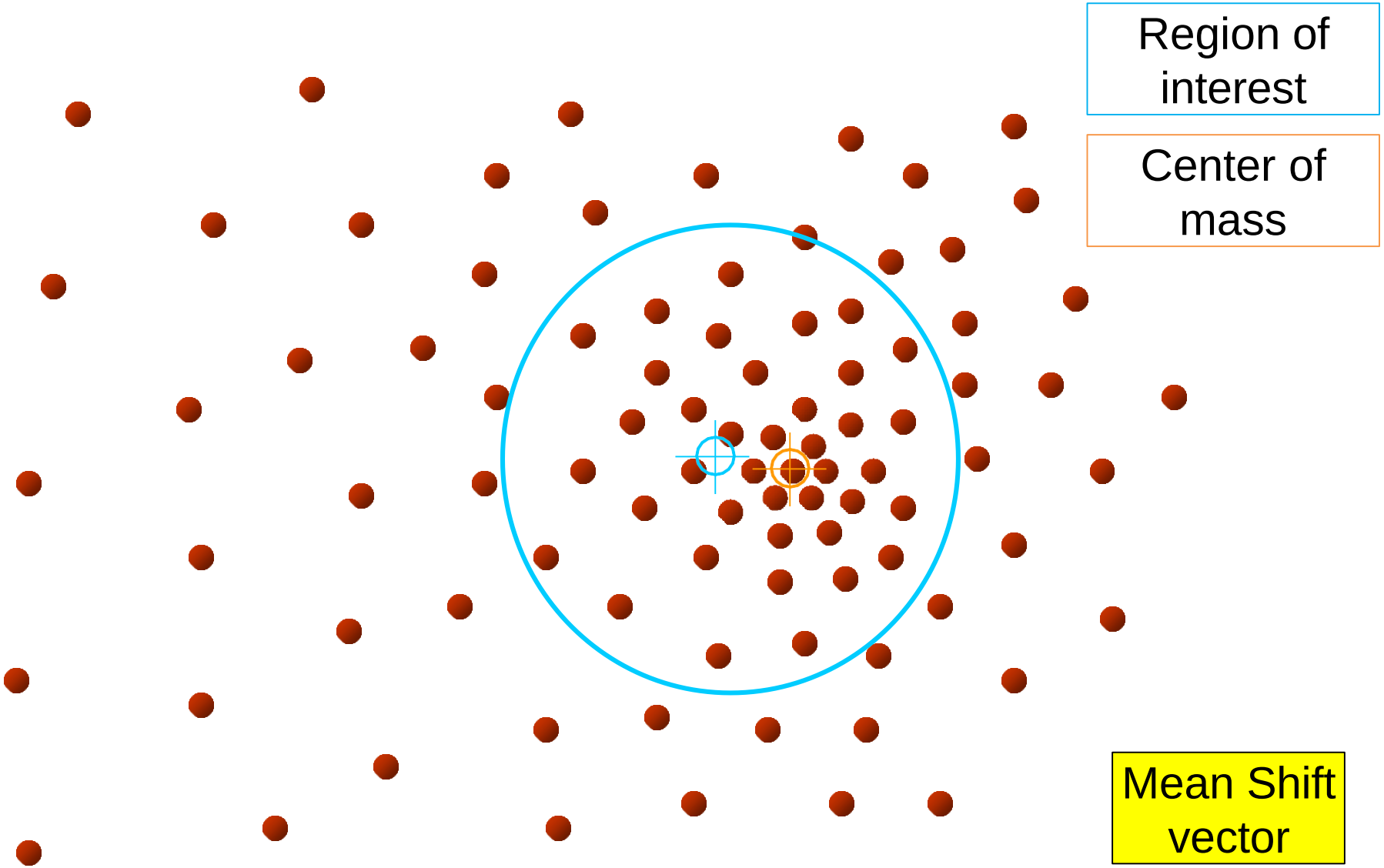
Region Splitting



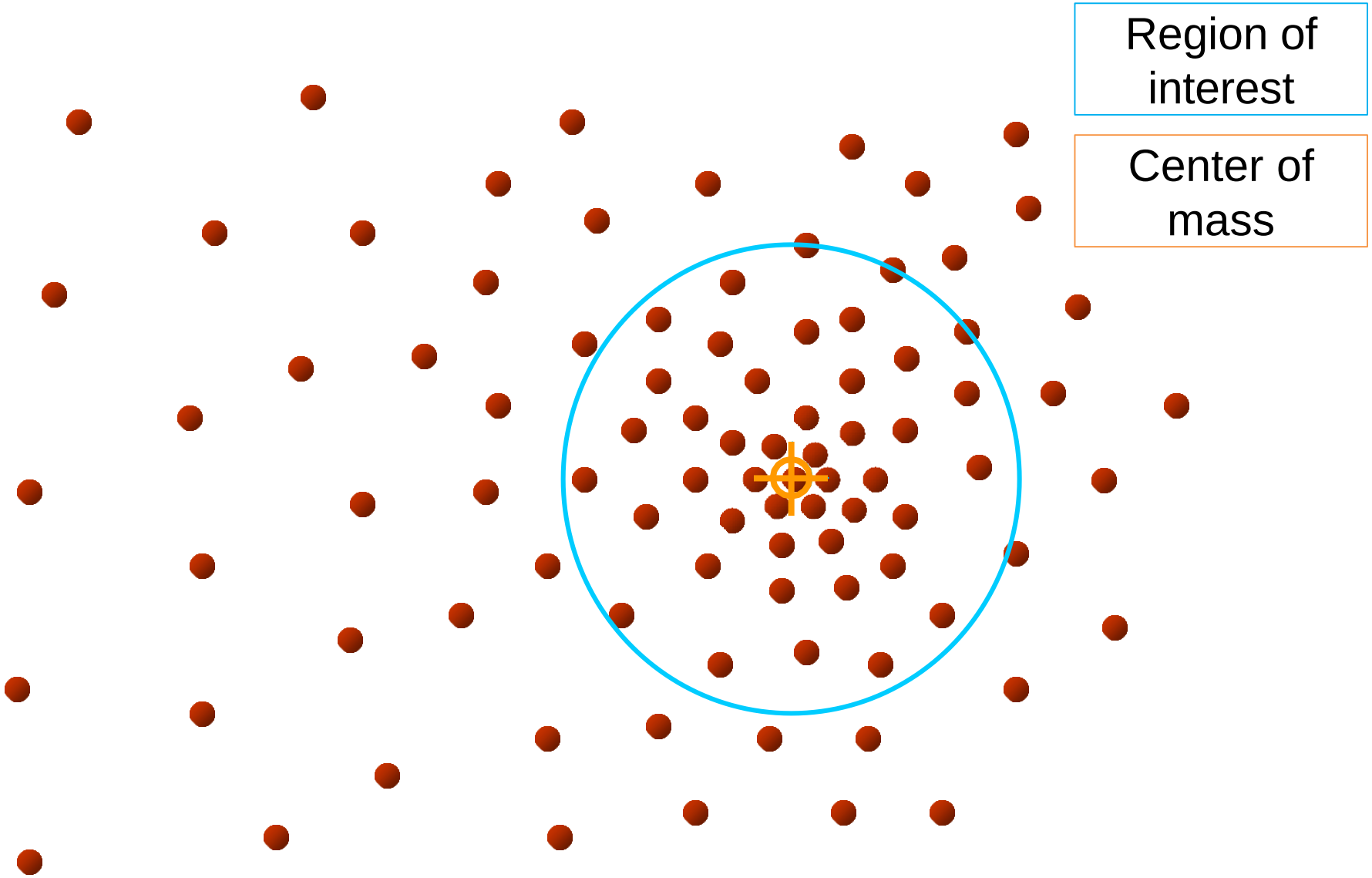
Region Splitting



Region Splitting



Region Splitting



Region Splitting

➢ Mean Shift Algorithm

Clustering:

1: Set mi = fi as initial mean for each pixel i.

2: Repeat the following for each mean mi:

a: Place the window of size W around mi.

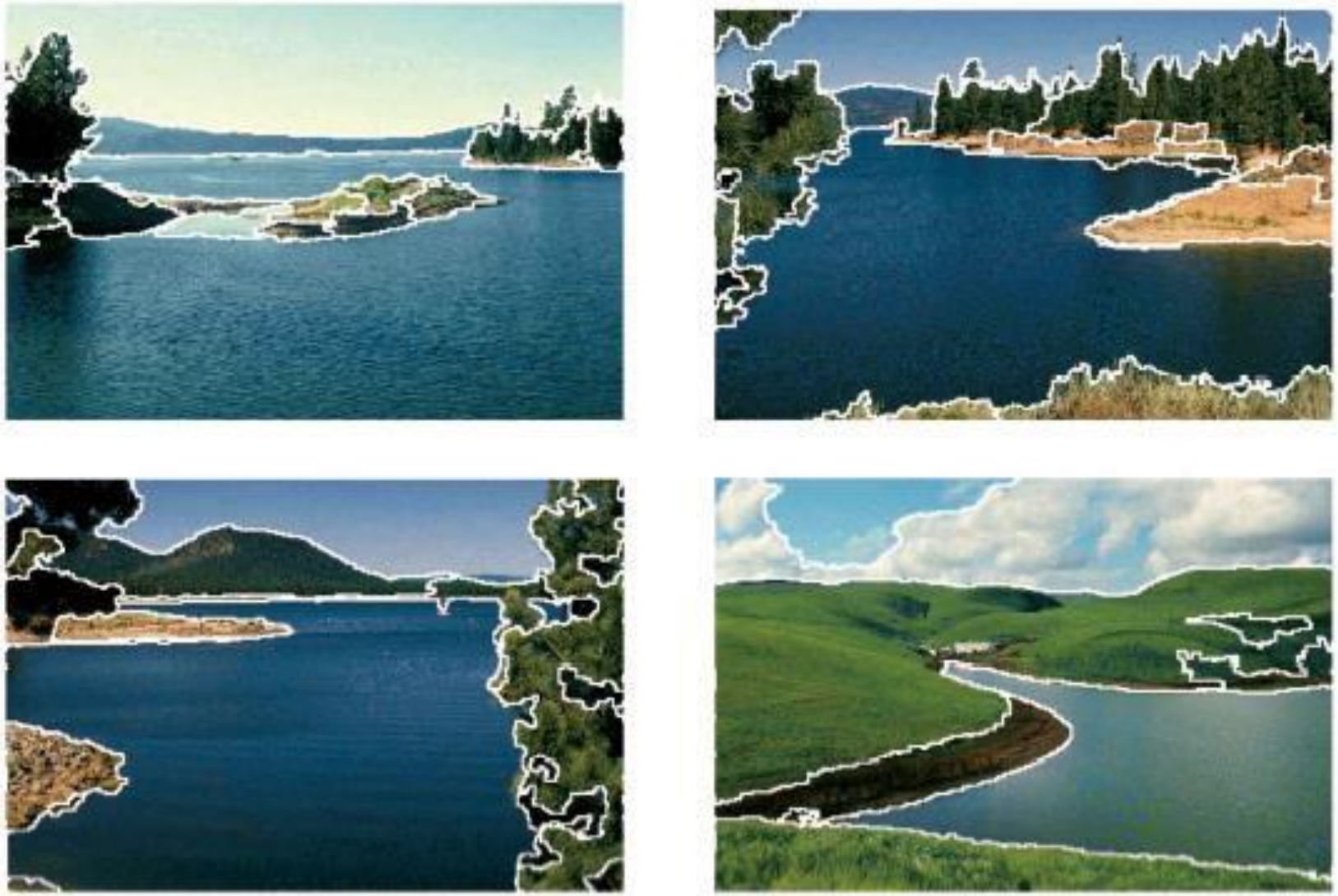
b: Compute centroid m within the window. Set mi = m.

c: Stop if shift in mean mi is less than a threshold then mi is the mode.

3: Label all pixels that have same mode as belonging to the same cluster.

Region Splitting

- **Example of Mean Shift Segmentation**

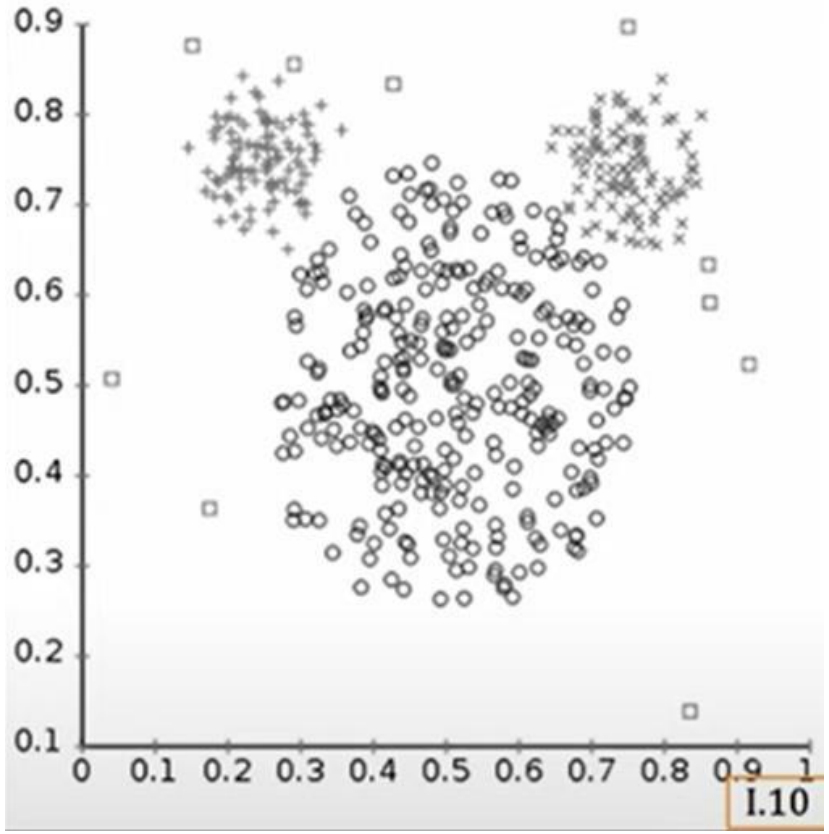


http://www.caip.rutgers.edu/~comanici/MSPAMI/msPamiResults.html

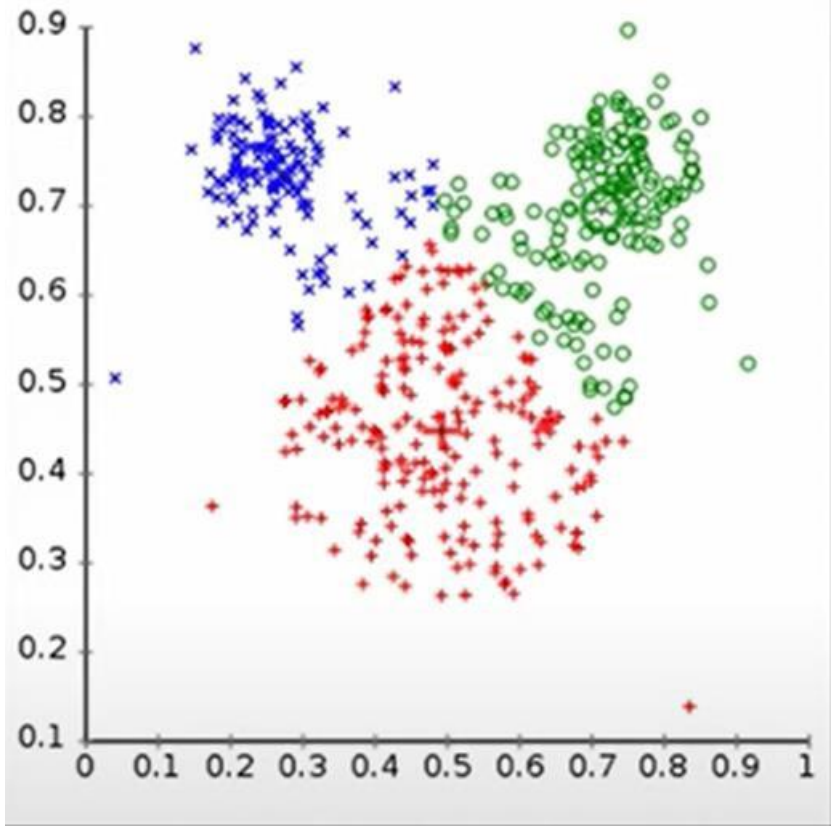
Region Splitting

➢ k-Means vs. Mean Shift

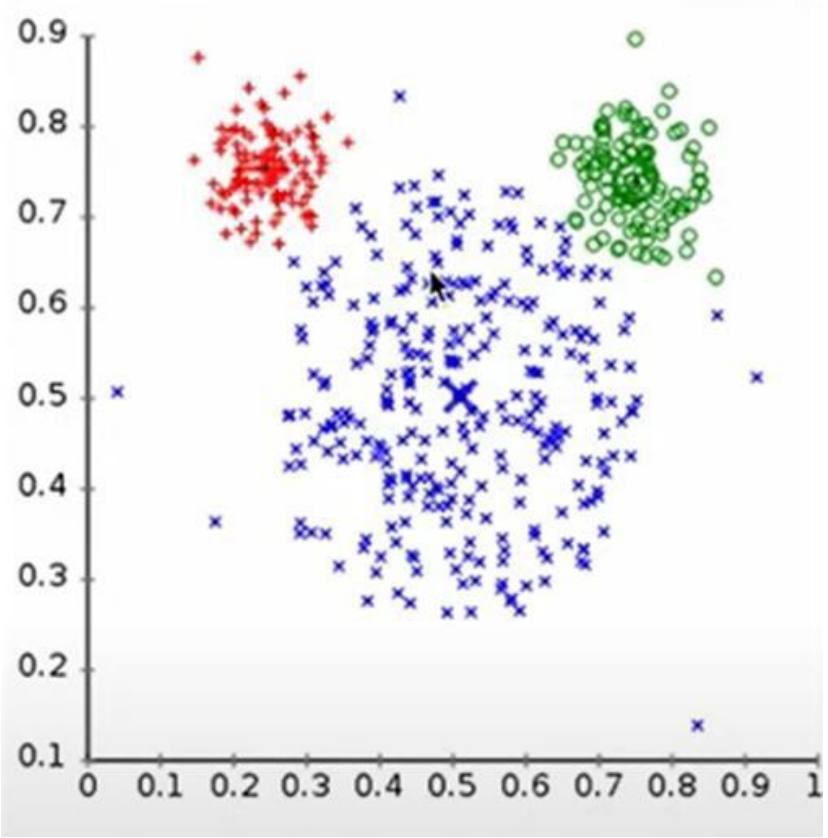
https://www.youtube.com/watch?v=PCNz_zttmtA



Original Data



K-Means (k=3)



Mean Shift

Region Splitting

- **k-Means vs. Mean Shift**

https://www.youtube.com/watch?v=PCNz_zttmtA

Original Data

K-Means (k=16)

Mean Shift

Region Splitting

- **Let's Summarize:**
  - Region Splitting
  - Region Merging